

Customer Shopping Behavior Analysis

A cross-sectional analysis of retail customer spend, discount behavior, and product performance

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Tools used	Python (pandas) · SQL · Power BI
Dataset	3,900 customers · 18 attributes
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1. Project Overview

This report covers an end-to-end analysis of a retail customer shopping dataset, working through data cleaning and feature engineering in Python, eight business questions answered in SQL, and a Power BI dashboard built on the results.

The dataset is a single snapshot of 3,900 customers — one row per customer, with no transaction timestamps. That rules out time-series or retention-style analysis; the questions in this report are scoped to comparing customer segments against each other at a point in time, rather than tracking behavior over time.

2. Dataset Summary

- **Rows:** 3,900 **Columns:** 18
- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method
- **Missing data:** 37 values (under 1%), all in Review Rating

3. Data Preparation (Python)

Before any business question could be answered, the raw data needed cleaning, validation, and a few new fields derived from what was already there.

3.1 Missing values

The 37 missing Review Rating values were imputed using the **median rating within each product category**, rather than one global median — ratings cluster differently by category, so a category-specific fill is a more defensible choice.

3.2 Redundant column check

Before treating `discount_applied` and `promo_code_used` as two independent signals, they were checked against each other directly. They turned out to be identical for every customer in the dataset — a promo code was used if and only if a discount was applied. The redundant column was dropped, and rather than losing an analysis angle, the project instead looked at how discount usage relates to subscription status and season (Sections 4.3 and 4.6).

3.3 Outlier check

Purchase Amount, Review Rating, and Previous Purchases were checked for outliers using the IQR method before any aggregation. None of the three fields had any values outside 1.5x the interquartile range — this is a clean dataset, and the averages used throughout the SQL analysis don't require trimming.

3.4 Feature engineering

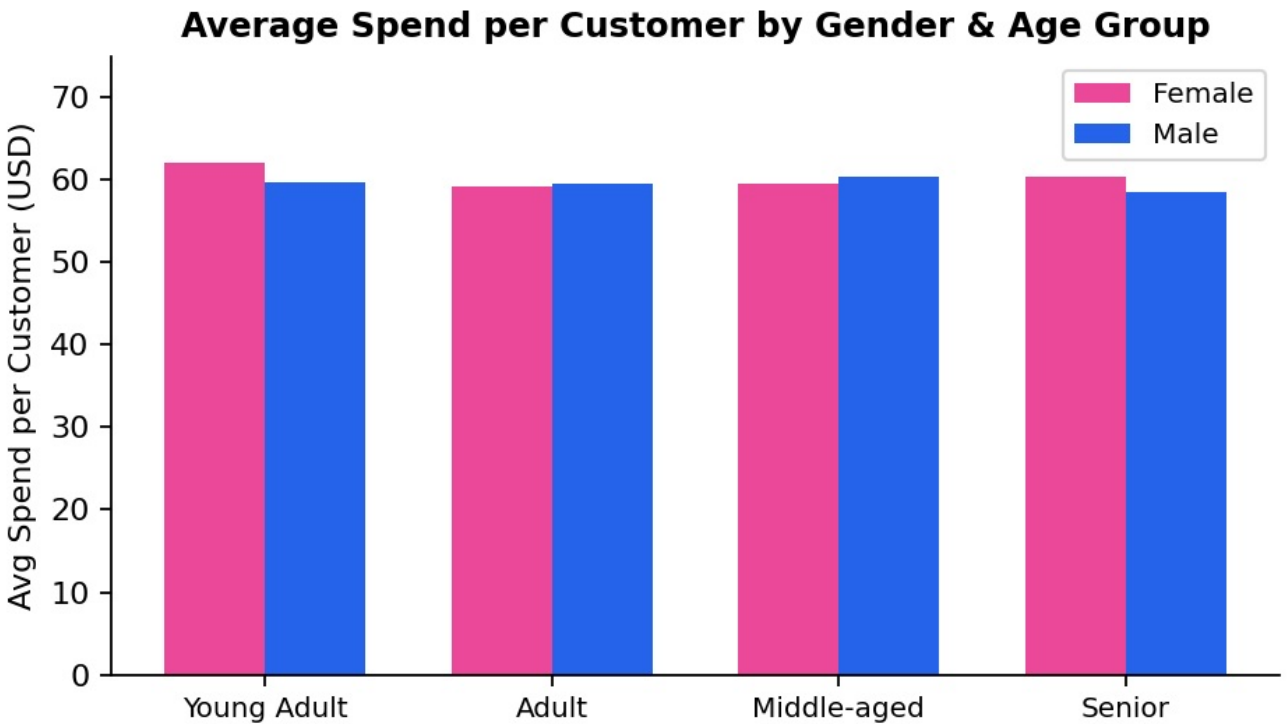
- **age_group** — ages split into quartiles (Young Adult, Adult, Middle-aged, Senior) rather than fixed-width bins, so each group holds a comparable number of customers.
- **purchase_frequency_days** — the text label in Frequency of Purchases (e.g. "Weekly", "Monthly") mapped to an approximate numeric day count, making it usable in quantitative comparisons.
- **spend_tier** — Purchase Amount split into Low / Mid / High using the dataset's own quartile boundaries, used in the customer value segmentation in Section 4.6.

4. Business Analysis (SQL)

Eight questions were used to look at customer spend, discount behavior, and product performance from a few different angles. Where a question involves comparing groups of different sizes (e.g. gender, age), both an average and a total are reported side by side — a larger group will always show a larger total even with identical average spend per person, so reporting only the total can be misleading.

4.1 Average spend per customer by gender and age group

Gender	Age Group	Customers	Avg Spend / Customer	Total Revenue
Female	Young Adult	325	\$61.98	\$20,142
Male	Middle-aged	653	\$60.32	\$39,391
Female	Senior	290	\$60.29	\$17,484
Male	Young Adult	703	\$59.75	\$42,001
Male	Adult	642	\$59.53	\$38,219
Female	Middle-aged	333	\$59.48	\$19,806
Female	Adult	300	\$59.20	\$17,759
Male	Senior	654	\$58.53	\$38,279



FINDING

Average spend per customer is essentially flat across every gender/age combination — all eight groups fall within a \$58–\$62 band. Total revenue differs mainly because group sizes differ (e.g. Male/Young Adult has more than twice the customers of Female/Adult), not because any demographic spends meaningfully more per person. Demographic targeting on its own is unlikely to be a strong lever for increasing spend in this dataset.

4.2 Average order value by payment method

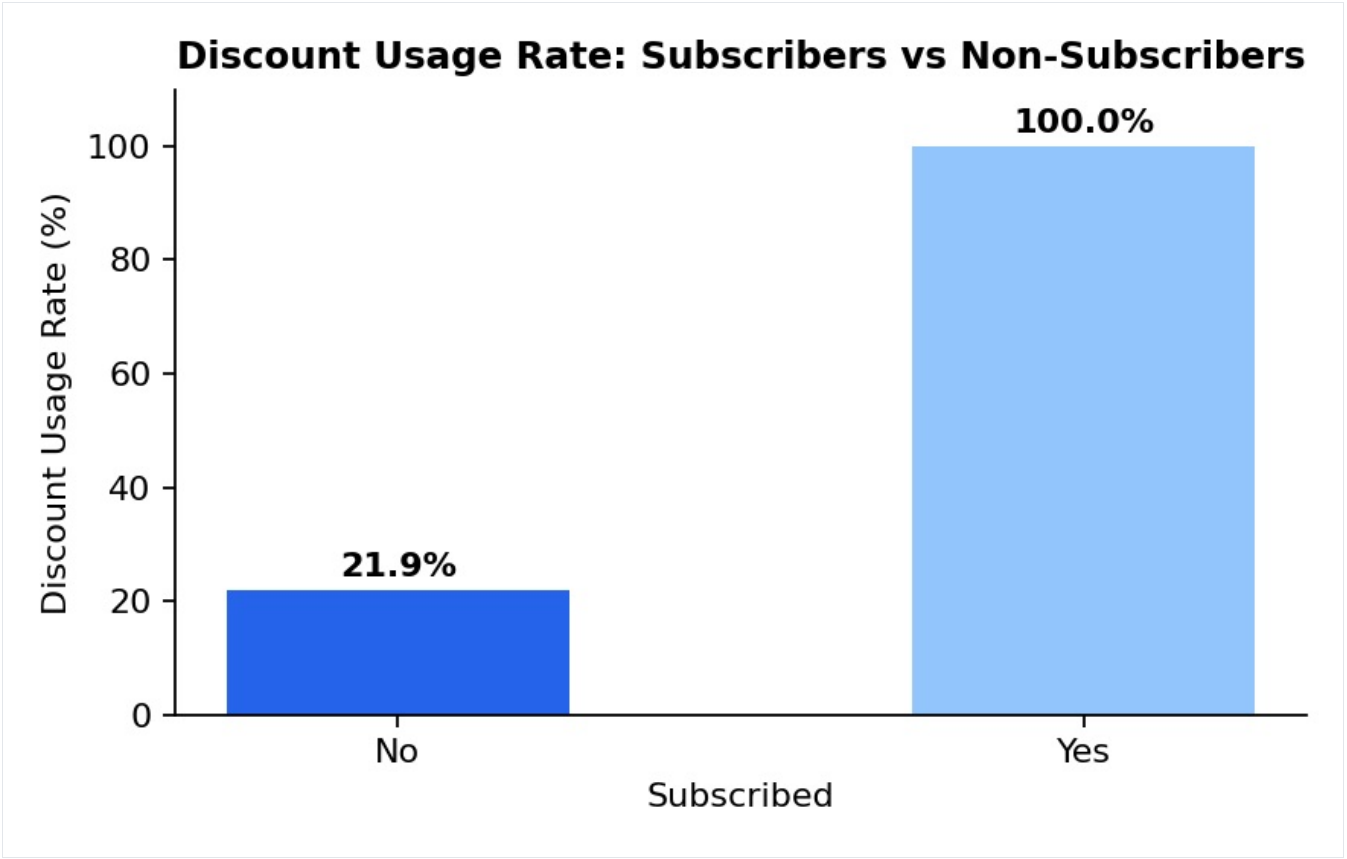
Payment Method	Total Orders	Avg Order Value
Debit Card	636	\$60.92
Credit Card	671	\$60.07
Bank Transfer	612	\$59.71
Cash	670	\$59.70
PayPal	677	\$59.25
Venmo	634	\$58.95

FINDING

Same pattern as Section 4.1 — payment method shows no meaningful relationship with order value (a \$2 spread across all six methods). Payment method appears to be a matter of customer preference rather than a marker of spend behavior.

4.3 Discount usage: subscribers vs. non-subscribers

Subscribed	Customers	Discount Usage Rate	Avg Spend
No	2,847	21.9%	\$59.87
Yes	1,053	100.0%	\$59.49



FINDING — THE STRONGEST SIGNAL IN THIS ANALYSIS

Every single subscribed customer in this dataset has a discount applied to their purchase, compared to roughly 1 in 5 non-subscribers. Average spend between the two groups is nearly identical (\$59.49 vs \$59.87), so subscribers aren't spending more — they're universally discounted. This raises a real question for the business: is the subscription program functioning as intended (a loyalty mechanism that earns its own engagement), or has it effectively become a 100%-discount-rate segment that isn't differentiated from a regular promotion? See Recommendation 1 in Section 7.

4.4 Lowest-rated category & size combinations

Category	Size	Orders	Avg Review Rating
Clothing	M	778	3.69
Outerwear	M	148	3.69
Footwear	M	267	3.71
Clothing	XL	194	3.73
Accessories	L	307	3.74

FINDING

Ratings are tightly clustered overall (roughly 3.69–3.86 across every category/size combination, filtered to combinations with at least 20 orders) — there's no single size or category with a clear quality problem. Medium-sized items across Clothing, Outerwear, and Footwear sit very slightly below the rest, but the gap is small enough that it's more useful as a thing to monitor than as an urgent fix.

4.5 Average spend by category and shipping type

Category	Shipping Type	Orders	Avg Spend
Outerwear	Store Pickup	53	\$66.94

Footwear	2-Day Shipping	90	\$62.14
Accessories	2-Day Shipping	206	\$61.99
Clothing	Free Shipping	294	\$61.99
...	...	...	...
Outerwear	Standard	49	\$52.06

FINDING

Outerwear shows the widest spend spread by shipping type of any category — Store Pickup customers spend nearly \$15 more on average than Standard shipping customers buying the same category. Every other category stays within a roughly \$4–5 band regardless of shipping choice. Outerwear is worth a closer look on its own; the other three categories don't show shipping-driven spend differences worth acting on.

4.6 Customer value tiers (purchase history × spend)

History Tier	Spend Tier	Customers
Mid History	Mid Spend	1,020
Low History	Mid Spend	515
Mid History	Low Spend	491
Mid History	High Spend	447
High History	Mid Spend	468
Low History	High Spend	250
Low History	Low Spend	249
High History	Low Spend	231
High History	High Spend	229

FINDING

Purchase history and spend tier don't move together — a customer with a long purchase history is about as likely to be a Low, Mid, or High spender as a customer with a short history. Quartile cuts (rather than the original project's single fixed threshold) were used for both tiers here, since previous_purchases is close to uniformly distributed across 1–50 in this dataset, making quartiles a fairer split than an arbitrary cutoff. The practical takeaway: "loyal" by purchase count and "valuable" by spend are two different customer properties here, and a segmentation strategy should track both rather than assuming one predicts the other.

4.7 Top 3 products per category, with rating

Category	Rank	Product	Orders	Avg Rating
Accessories	1	Jewelry	171	3.76
Accessories	2	Sunglasses	161	3.75
Clothing	1	Pants	171	3.72
Footwear	1	Sandals	160	3.84
Outerwear	1	Jacket	163	3.76

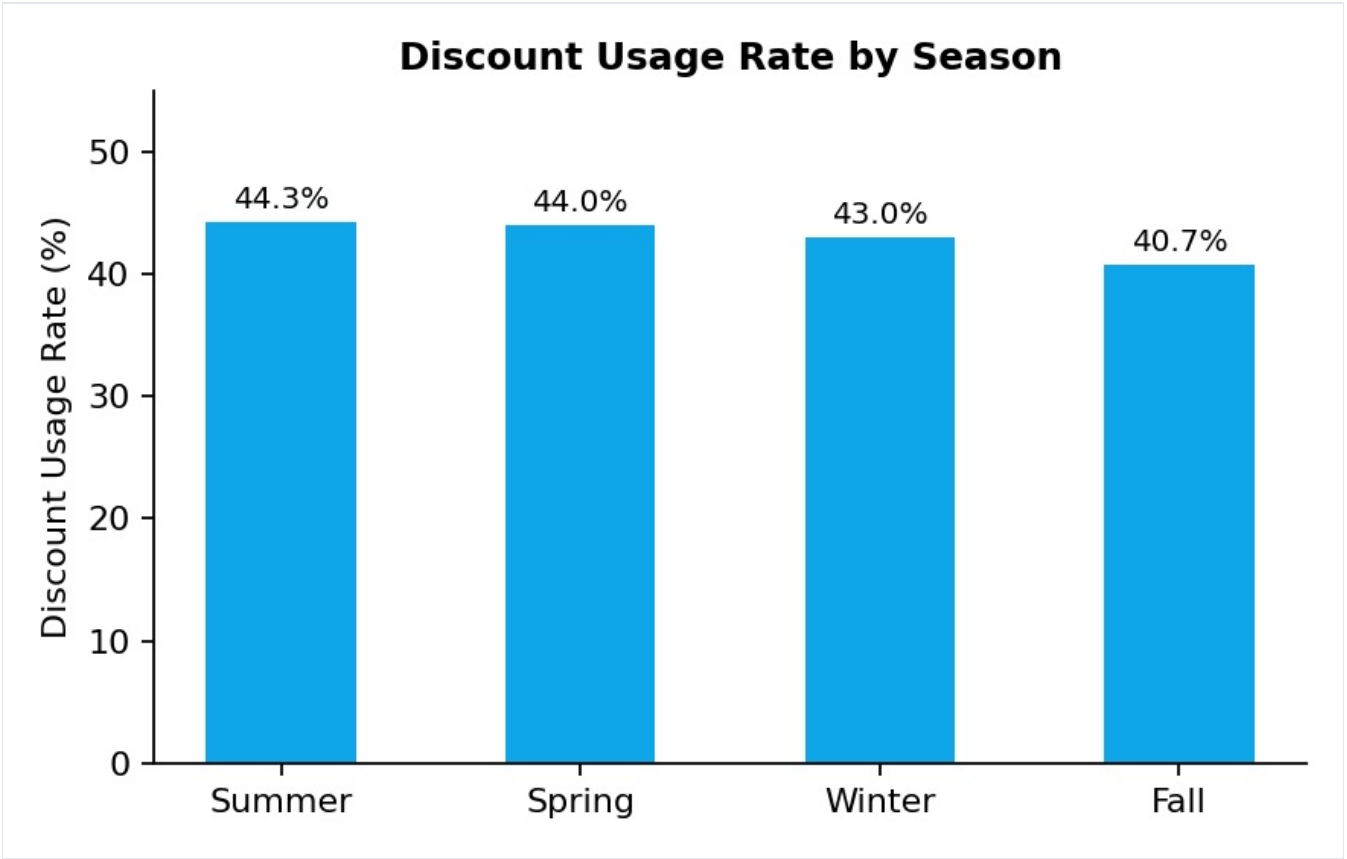
FINDING

Pairing order volume with rating (rather than reporting volume alone) shows that the most purchased item in each category is also reasonably well-rated — there's no case here of a high-volume product carrying a low rating, which would be a red flag worth investigating. Sandals stand out slightly as the best-rated top-seller across all four categories.

4.8 Discount usage by season

Season	Orders	Discount Usage Rate
Summer	955	44.3%
Spring	999	44.0%

Winter	971	43.0%
Fall	975	40.7%



FINDING

Discount usage is fairly stable across seasons (a roughly 3.6-point spread between the highest and lowest), with a slight dip in Fall. This isn't a strong seasonal pattern — worth noting as a "checked, no major effect found" result rather than overstating a 3-point difference as a trend.

5. Power BI Dashboard

An interactive Power BI dashboard was built on top of the SQL output above, with filters for subscription status, gender, category, and shipping type, so the findings in Section 4 can be explored interactively rather than only read as static tables.

[Insert dashboard screenshot here once built]

6. Limitations

Worth stating plainly: this dataset is a single snapshot with no transaction dates, so none of the findings above describe trends over time, seasonality of actual purchase dates, or customer retention. `previous_purchases` and `frequency_of_purchases` are the closest available proxies for behavior over time, but both are static counts/labels rather than a timestamped history, and should be read as such.

7. Business Recommendations

- 1. Re-examine the subscription program's value proposition**

Every subscriber receives a discount, but subscribers don't spend more on average than non-subscribers who self-select into a discount. If the goal of the subscription program is to build loyalty beyond price, it currently isn't measurably doing that in this data — worth testing non-price subscriber perks (early access, free shipping tiers, exclusive items) and tracking whether that shifts spend, rather than assuming the 100% discount rate alone is driving the relationship.
- 2. Don't lead with demographic targeting**

Gender, age group, and payment method all show flat average spend per customer. Marketing spend aimed at "high-value demographics" isn't well supported by this data — segment by behavior (Section 4.6's history/spend tiers) instead of who the customer is.

3. Investigate the Outerwear shipping/spend gap

Outerwear customers who choose Store Pickup spend close to \$15 more on average than those using Standard shipping — the widest gap of any category. Worth a follow-up look at whether this reflects a specific high-value Outerwear subsegment (e.g. heavier/pricier coats bought locally) before generalizing this into a marketing decision.

4. Track purchase history and spend as two separate segments

A customer with a long purchase history isn't reliably a high spender, and vice versa. Loyalty and value programs that assume "more purchases = more valuable" should be reviewed against this — the two need separate criteria.

5. No urgent action needed on ratings or seasonal discounting

Review ratings are tightly clustered with no clear quality outlier, and discount usage by season shows only a minor Fall dip. Both are worth monitoring on an ongoing basis rather than acting on immediately — flagging a non-finding honestly is as useful as flagging a real one.